

This section provides learners with a knowledge and understanding of the structure and functions of the transport and gas exchange systems in mammals and terrestrial plants.

Learners will gain an appreciation of the need for mass transport systems and the significance of the surface area to volume ratio in multicellular organisms. The mass flow of materials within mammals and the exchange of molecules are studied in the context of the need to maintain a high metabolic rate.

Technological advances enable the activity of the heart, circulation and gas exchange systems to be monitored. Learners will be able to apply their knowledge of physiology to the interpretation of data on vital measurements of heart function, blood pressure and pulmonary ventilation. The importance of timely emergency treatment in both cardiac and respiratory arrest is also studied.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.